

8-1-2021

Correcting Interval-Valued Expert Estimates: Empirical Formulas Explained

Laura A. Berrout Ramos

The University of Texas at El Paso, laberroutra@utep.edu

Vladik Kreinovich

The University of Texas at El Paso, vladik@utep.edu

Kittawit Autchariyapanitkul

Maejo University, kittawit_a@mju.ac.th

Follow this and additional works at: https://scholarworks.utep.edu/cs_techrep



Part of the [Computer Sciences Commons](#), and the [Mathematics Commons](#)

Comments:

Technical Report: UTEP-CS-21-72

Recommended Citation

Berrout Ramos, Laura A.; Kreinovich, Vladik; and Autchariyapanitkul, Kittawit, "Correcting Interval-Valued Expert Estimates: Empirical Formulas Explained" (2021). *Departmental Technical Reports (CS)*. 1605.
https://scholarworks.utep.edu/cs_techrep/1605

This Article is brought to you for free and open access by the Computer Science at ScholarWorks@UTEP. It has been accepted for inclusion in Departmental Technical Reports (CS) by an authorized administrator of ScholarWorks@UTEP. For more information, please contact lweber@utep.edu.

Correcting Interval-Valued Expert Estimates: Empirical Formulas Explained

Laura A. Berrou Ramos, Vladik Kreinovich, and
Kittawit Autchariyapanitkul

Abstract Experts' estimates are approximate. To make decisions based on these estimates, we need to know how accurate these estimates are. Sometimes, experts themselves estimate the accuracy of their estimates – by providing the interval of possible values instead of a single number. In other cases, we can gauge the accuracy of the experts' estimates by asking several experts to estimate the same quantity and using the interval range of these values. In both situations, sometimes the interval is too narrow – e.g., if an expert is overconfident. Sometimes, the interval is too wide – if the expert is too cautious. In such situations, we correct these intervals, by making them, correspondingly, wider or narrower. Empirical studies show that people use specific formulas for such corrections. In this paper, we provide a theoretical explanation for these empirical formulas.

1 Formulation of the Problem

Expert estimates are needed. In economics – as in many other application areas – we often rely on expert estimates.

Expert estimates are approximate. Of course, expert estimates are approximate. So, to be able to use them effectively, we need to know how accurate they are.

How do we usually gauge the accuracy of expert estimates. There are two main ideas on how to gauge the accuracy of an expert estimate:

- the first is to ask the expert him/herself to gauge this accuracy, namely, to provide an *interval* of possible values instead of a single value;

Laura A. Berrou Ramos and Vladik Kreinovich
University of Texas at El Paso, 500 W. University, El Paso, TX 79968, USA
e-mail: laberroutra@utep.edu, vladik@utep.edu

Kittawit Autchariyapanitkul
Faculty of Economics, Maejo University, Chiang Mai, Thailand, e-mail: Kittawit.a@mju.ac.th

- the second is to ask one or more other experts, and to consider the range formed by their values as the reasonable range of possible values for the corresponding quantity; for example, if one expert predicts the value 15, and the two others predict 10 and 20, then we take the interval $[10, 20]$.

Need for correction. When an expert provides an interval of possible values:

- sometimes, the expert is too confident, and the interval provided by the expert is too narrow; in this case, a natural idea is to widen it;
- sometimes, the expert is too cautious, and the interval provided by the expert is too wide; in this case, a natural idea is to make it narrower.

Similarly, when we have an interval range formed by numerical estimates made by different experts:

- sometimes, the experts' estimates are too close to each other, so the resulting interval is too narrow; in this case, a natural idea is to widen it;
- sometimes, the experts' estimates are too far away from each other; for example, one expert's predictions are too optimistic, and another expert's predictions are too pessimistic; in this case, the resulting interval is too wide, so a natural idea is to make it narrower.

How do people correct the corresponding intervals? Empirical data shows the corrected version $[A, B]$ of the original interval $[a, b]$ usually follows the formula

$$[A, B] = \left[a \cdot \frac{1 + \alpha}{2} + b \cdot \frac{1 - \alpha}{2}, a \cdot \frac{1 - \alpha}{2} + b \cdot \frac{1 + \alpha}{2} \right] \quad (1)$$

for some value $\alpha > 0$;

- the values $\alpha < 1$ correspond to shrinking, while
- the values $\alpha > 1$ correspond to stretching of the interval;

see, e.g., [3] and references therein.

Historical comment. The formula (1) was first proposed in [1] for expert estimates of probabilities. In [2], this formula was extended to general (not necessarily probabilistic) expert estimates.

Remaining problem. Why do people use this particular formula to correct interval-valued expert estimates?

In this paper, we provide a possible explanation for this formula.

2 Analysis of the Problem

What we look for. We are looking for a formula that describes the corrected interval $[A, B]$ – i.e., that describes both endpoints A and B of this interval – in terms of the

inputs interval $[a, b]$ – i.e., in terms of its endpoints a and b . In other words, we need to find algorithms $A(a, b)$ and $B(a, b)$ that, given the expert-provided values a and b , produced the corrected values A and B .

Let us analyze what are the natural properties of these algorithms.

The correction formula should not depend on the monetary unit. The same financial predictions can be described in different monetary units. For example, predictions related to Mexican economy can be made in Mexican pesos or in US dollars.

We are looking for general correction formulas, formulas that would be applicable to all possible interval-valued expert estimates. Suppose that we first applied this formula to the interval $[a, b]$ described in one monetary units. Then, in these units, the corrected interval takes the form

$$[A(a, b), B(a, b)]. \quad (2)$$

Another possibility is:

- to first translate into a different monetary unit,
- to make a correction there, and then
- to translate the result back into the original monetary unit.

If we select a different monetary unit which is λ times smaller than the original one, then all numerical values multiply by λ . In particular, in the new units, the original interval $[a, b]$ will take the form $[\lambda \cdot a, \lambda \cdot b]$. If we apply the same correction algorithm to this interval, we get – in the new units – the following corrected interval:

$$[A(\lambda \cdot a, \lambda \cdot b), B(\lambda \cdot a, \lambda \cdot b)]. \quad (3)$$

To describe the corrected interval (3) in the original units, we need to divide both its endpoints by λ . As a result, we get the corrected interval expressed in the original units:

$$\left[\frac{1}{\lambda} \cdot A(\lambda \cdot a, \lambda \cdot b), \frac{1}{\lambda} \cdot B(\lambda \cdot a, \lambda \cdot b) \right]. \quad (4)$$

As we have mentioned, it is reasonable to require that the corrected interval should be the same whether we use the original monetary units or different units, i.e., that we should have

$$[A(a, b), B(a, b)] = \left[\frac{1}{\lambda} \cdot A(\lambda \cdot a, \lambda \cdot b), \frac{1}{\lambda} \cdot B(\lambda \cdot a, \lambda \cdot b) \right]. \quad (5)$$

The two intervals are equal if their left endpoints are equal to each other and their right endpoints are equal to each other. So, the equality (5) means that:

$$A(a, b) = \frac{1}{\lambda} \cdot A(\lambda \cdot a, \lambda \cdot b) \quad (6)$$

and

$$B(a, b) = \frac{1}{\lambda} \cdot B(\lambda \cdot a, \lambda \cdot b). \quad (7)$$

Terminological comment. The properties (6)–(7) describing the fact that the transformation

$$[a, b] \mapsto [A(a, b), B(a, b)]$$

does not change if we change the measuring units (i.e., *re-scale* all the numerical values) is known as *scale-invariance*.

Shift-invariance. Suppose that the expected company's income consists of:

- the fixed amount f – e.g., determined by the current contracts – and
- some additional amount x that will depend on the relation between supply and demand.

Suppose that the expert predicts this additional amount to be somewhere in the interval $[a, b]$. This means that the overall company's income is predicted to be between $f + a$ and $f + b$, i.e., somewhere in the interval $[f + a, f + b]$.

If we believe that the expert estimate needs corrections, then we have two possible ways to perform this correction:

- we can apply the correction to the original interval $[a, b]$, resulting in the corrected interval estimate $[A(a, b), B(a, b)]$ for the additional income; in this case, the interval estimate for the overall income will be

$$[f + A(a, b), f + B(a, b)]; \quad (8)$$

- alternatively, we can apply the correction to the interval $[f + a, f + b]$ describing the overall income; in this case, the resulting corrected interval for the overall income will have the form

$$[A(f + a, f + b), B(f + a, f + b)]. \quad (9)$$

It is reasonable to require that the two methods should lead to the exact same interval estimate for the overall income:

$$[f + A(a, b), f + B(a, b)] = [A(f + a, f + b), B(f + a, f + b)], \quad (10)$$

i.e., equivalently,

$$f + A(a, b) = A(f + a, f + b) \quad (11)$$

and

$$f + B(a, b) = B(f + a, f + b). \quad (12)$$

Terminological comment. The properties (11)–(12) describing the fact that the transformation

$$[a, b] \mapsto [A(a, b), B(a, b)]$$

does not change if we shift all the inputs by the same amount f is known as *shift-invariance*.

Sign invariance. One of the possible expert predictions is, e.g., how much bank B_1 will owe a bank B_2 at a certain future date. This amount can be positive – meaning that the bank B_1 will owe some money to the bank B_2 . This amount can also be negative – meaning that, according to the expert, at the given future date, the bank B_2 will owe money to the bank B_1 .

Suppose that the expert estimates this amount by an interval $[a, b]$. This means that if we ask the same expert a different question: how much money will the bank B_2 owe to the bank B_1 – this expert will provide the interval $[-b, -a]$, i.e., the set of all the values $-x$ when $x \in [a, b]$.

In this case, we also have two possible ways to perform this correction:

- we can apply the correction to the original interval $[a, b]$, resulting in the corrected interval estimate

$$[A(a, b), B(a, b)]; \quad (13)$$

- alternatively, we can apply the correction to the interval $[-b, -a]$ describing how much the bank B_2 will owe to the bank B_1 , and get the corrected interval $[A(-b, -a), B(-b, -a)]$ for this amount; by changing the sign, we get an interval estimate of how much the bank B_1 will owe to the bank B_2 :

$$[-B(-b, -a), -A(-b, -a)]. \quad (14)$$

It is reasonable to require that the two methods should lead to the exact same interval estimate for the overall amount:

$$[A(a, b), B(a, b)] = [-B(-b, -a), -A(-b, -a)], \quad (15)$$

i.e., equivalently,

$$A(a, b) = -B(-b, -a) \quad (16)$$

and

$$B(a, b) = -A(-b, -a). \quad (17)$$

Terminological comment. We will call the properties (16)–(17) describing the fact that the transformation

$$[a, b] \mapsto [A(a, b), B(a, b)]$$

does not change if change all the signs, *sign-invariance*.

3 Definitions and the Main Result

Definition. We say that a mapping $[a, b] \mapsto [A(a, b), B(a, b)]$ is:

- scale-invariant if it satisfies the properties (6) and (7) for all $a < b$ and for all $s\lambda > 0$;
- scale-invariant if it satisfies the properties (11) and (12) for all $a < b$ and f ; and
- sign-invariant if it satisfies the properties (16) and (17) for all $a < b$.

Comment. One can easily check that the transformation (1) is scale-, shift-, and sign-invariant. It turns out that every scale-, shift-, and sign-invariant transformation has the form (1). This explains why people use such transformations to correct interval-valued expert estimates.

Proposition. Every scale-, shift-, and sign-invariant transformation has the form (1).

Proof. Indeed, due to shift-invariance for $f = a$, we have

$$B(a, b) = a + B(0, b - a). \quad (18)$$

Due to scale-invariance for $\lambda = b - a$, we have

$$B(0, 1) = \frac{1}{b - a} \cdot B(0, b - a),$$

hence

$$B(0, b - a) = (b - a) \cdot B(0, 1). \quad (19)$$

Let us denote

$$\alpha \stackrel{\text{def}}{=} 2B(0, 1) - 1,$$

then

$$B(0, 1) = \frac{1 + \alpha}{2},$$

and the formula (19) takes the form

$$B(0, b - a) = (b - a) \cdot \frac{1 + \alpha}{2} = b \cdot \frac{1 + \alpha}{2} - a \cdot \frac{1 + \alpha}{2}. \quad (20)$$

Substituting the expression (20) into the formula (18), we get

$$B(a, b) = a + b \cdot \frac{1 + \alpha}{2} - a \cdot \frac{1 + \alpha}{2} = a \cdot \frac{1 - \alpha}{2} + b \cdot \frac{1 + \alpha}{2}.$$

This is exactly the expression for $B(a, b)$ corresponding to the formula (1).

Now, by using sign-invariance, namely, the formula (16), we conclude that

$$A(a, b) = -B(-b, -a) = -\left((-b) \cdot \frac{1 - \alpha}{2} + (-a) \cdot \frac{1 + \alpha}{2}\right) =$$

$$a \cdot \frac{1 + \alpha}{2} + b \cdot \frac{1 - \alpha}{2}.$$

This is also exactly the expression for $A(a, b)$ corresponding to the formula (1). Thus, the proposition is proven.

Comment. In addition to empirically confirming the formula (1), the paper [3] also mentions an additional empirical fact that it find difficult to explain: that if two experts provide interval estimates, people put trust into pairs of intervals that have comparable width.

From our viewpoint, this is easy to explain: if two experts, based on the same data, provide completely different estimates of how accurately we can make a prediction based on this data, then we do not trust any of these experts. This is one of the cases when the two supposed experts are inconsistent. A similar phenomenon happens when the two experts provide drastically different numerical estimates: in this case, we do not trust either of them.

Acknowledgments

This work was supported in part by the National Science Foundation grants 1623190 (A Model of Change for Preparing a New Generation for Professional Practice in Computer Science), and HRD-1834620 and HRD-2034030 (CAHSI Includes), and by the AT&T Fellowship in Information Technology.

It was also supported by the program of the development of the Scientific-Educational Mathematical Center of Volga Federal District No. 075-02-2020-1478, and by a grant from the Hungarian National Research, Development and Innovation Office (NRDI).

References

1. T. Gajdos and J.-C. Vergnaud, “Decisions with conflicting and imprecise information”, *Social Choice and Welfare*, 2013, Vol. 41, pp. 427–452, doi: 10.1007/s00355-012-0691-1
2. M. Smithson, “Conflict and ambiguity: preliminary models and empirical tests”, *Proceedings of the Eighth International Symposium on Imprecise Probability: Theories and Applications ISIPTA’2013*, Compiègne, France, July 2–5, 2013, pp. 303–310.
3. M. Smithson, “Probability judgments under ambiguity and conflict”, *Frontiers in Psychology*, 2015, Vol. 6, Paper 674, doi: 10.3389/fpsyg.2015.00674